

VEGF / VEGFR: 13 molecules stopped early

20 stopped trials across 13 molecules: Cabozantinib, Lenvatinib, Bevacizumab, Catequentinib and 9 more. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

3.1%

20 of 655 closed trials (95% CI 2.0%–4.7%)

13 / 5 / 0

efficacy / safety / both (adds to 20)

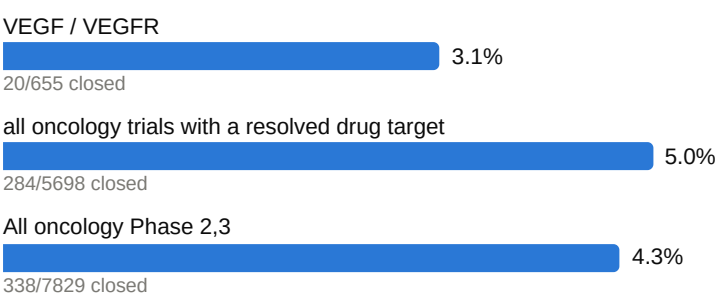
30.2%

of 2172 trials have closed · 1517 still open or unresolved

0.9%

stopped for a biological reason within 3 years of starting (95% CI 0.5%–1.6%; 774 still at risk; 1.5% area-wide)

How this compares



What is in this class: 471 distinct experimental drugs across 2172 trials (small molecule 1849, antibody 1286, unknown 364, protein 76). Most tested: BEVACIZUMAB (474), RIVOCERANIB (343), LENVATINIB (267), CATEQUENTINIB (259), PEMBROLIZUMAB (159). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 20 stops came from 20 sponsor-asset programmes across 18 sponsors; the largest (Yale University) contributed 1; removing that programme's trials from both sides leaves 2.9%. Screened alongside 85 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 38 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 8.9%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	7	241	2.9%	1.4%–5.9%
2018–2020	10	250	4.0%	2.2%–7.2%
2021–2024	3	164	1.8%	0.6%–5.2%

Where the stops sit

Efficacy / futility: 13, Safety: 5, Benefit–risk: 2.
Sponsors with most stops: Jiangsu Alphamab Biopharmaceuticals Co., Ltd (2), Sun Yat-sen University (2), Yale University (1), Great Novel Therapeutics Biotech & Medicals Corporation (1), Niguarda Hospital (1), Eli Lilly (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04955743	2	Yale University	PEMBROLIZUMAB, LENVATINIB	2025-05	Efficacy	Lack of activity in the drugs
NCT05770882	2	Great Novel Therapeutics Biotech & Medicals Corporation	TUCIDINOSTAT, REGORAFENIB	2025-01	Benefit–risk	The overall profile does not support development for Hepatocellular Carcinoma
NCT03686345	2	Niguarda Hospital	MIDOSTAURIN	2023-06	Efficacy	didn't reach the primary endpoint
NCT04145700	2	Eli Lilly LLY	RAMUCIRUMAB	2023-02	Efficacy	Study terminated for meeting protocol specified futility criteria.
NCT05001724	3	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, LENVATINIB	2023-02	Efficacy	Because the efficacy data did not meet expectations, the sponsor decided to terminate the study.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04438902	2	First Affiliated Hospital of Zhejiang University	OSIMERTINIB, CATEQUENTINIB	2022-12	Safety	Many patients can't tolerate the combination treatment due to adverse events.

The 6 most recent of 20 stopped trials. Every molecule behind all 20 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 30.2% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

Any mechanism, sponsor or indication, updated weekly. The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: clinicaltrialfailures.com/pricing

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxCClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.